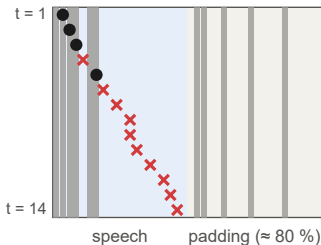


(a) One-shot retention

H2O, SnapKV, PyramidKV, PadSink-KV

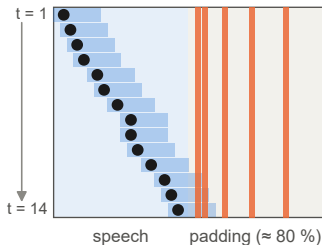
encoder positions →



(b) PadSink-Track

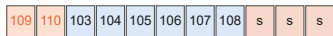
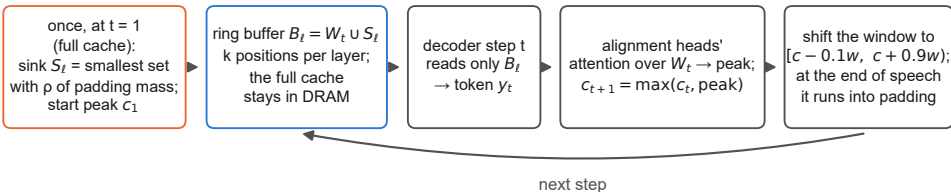
the selection is renewed at every step

encoder positions →



- position the decoder needs at step t
- ✗ needed but evicted → errors, loops
- one-shot set: chosen at $t = 1$, fixed
- window W_t : follows the alignment peak
- padding sink S_t : exact K/V, fixed

(c) One decoding step of PadSink-Track



ring buffer of one layer: the slots of positions 101, 102 that left the window are overwritten with the entering 109, 110 (0–3 slots per step, no per-step gather); s = sink slots. Attention ignores slot order, so the result equals reading the positions in order.